

Group 4

A Persona Based Approach for Exploring Architecturally Significant Requirements in Agile Projects

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Different Methods to derive ASR's

– Win Win

- The system is based on a negotiated deal between customer demands and cost and time to market
- Not quite suitable in a agile process since all is determined up front.
- Reference <http://www.1000sourcecodes.com/2012/05/software-engineering-winwin-spiral.html>

– i*

- Handles different, often competing, goals that depend on each other to undertake their tasks and achieve these goals. It covers both actor-oriented and [Goal modeling](#). i* models answer the question WHO and WHY, not what.
- Reference http://en.wikipedia.org/wiki/I*

– ASP-Lite

- Invented for use in agile projects – having a flexible and lightweight approach to architectural decision making.

Introduction ASP Lite

- Aims at fully explore and understand the impact of early architectural decisions using personas
- Focus primarily on quality concerns and constrains of the system – to identify ASR's
- Is used to design and evaluate the architectural design.
- Supports incremental delivery of the architectural design – supporting agile developement.

Architecturally Savvy Persona

- Is a realistic and engaging representation of a distinguishable specific stakeholder group
- Typically depicted through a picture and a personal description – including psyche, background, emotions and attitudes.
- Creating a persona involves surveying and interviewing stakeholders
- Typically from 5-8 Personas in a project

ASP-Lite

Identify Personas

– Theory

- Define Personas through Workshops & Brainstorms for Stakeholder Groups
- Stakeholder Groups defines User Stories

Creation of a Persona in ASP-lite Part 1

Persona picture,
name tag, and role.

Personalized background
details



Tom:
Age: 59, Professor

Tom is a long time traceability researcher. He has published numerous papers that have focused on tracing from source code to design and requirements. He has focused on using LDA, LSI, and various probabilistic approaches. He has also developed algorithms for visualizing the results of his traces.

Tom prefers coding in C++ on Linux. He plans to contribute components to the TRACY project; however he already has an established traceability research environment and therefore may not use all the TRACY features himself.

My user stories:

1. I need to be able to write components in C++ and integrate them easily into TraceLab experiments.
2. Experiments that I run using TraceLab must not take about the same amount of time to run as my existing experiments.
3. I need to be able to run TraceLab on Linux.
4. I need accessibility to benchmarks so I can compare new algorithms and techniques against previous results.
5. I need access to datasets with existing trace matrices.

My anti-stories:

1. I won't use TraceLab if it is buggy and keeps breaking.

Persona-related
user stories i.e.
win scenarios

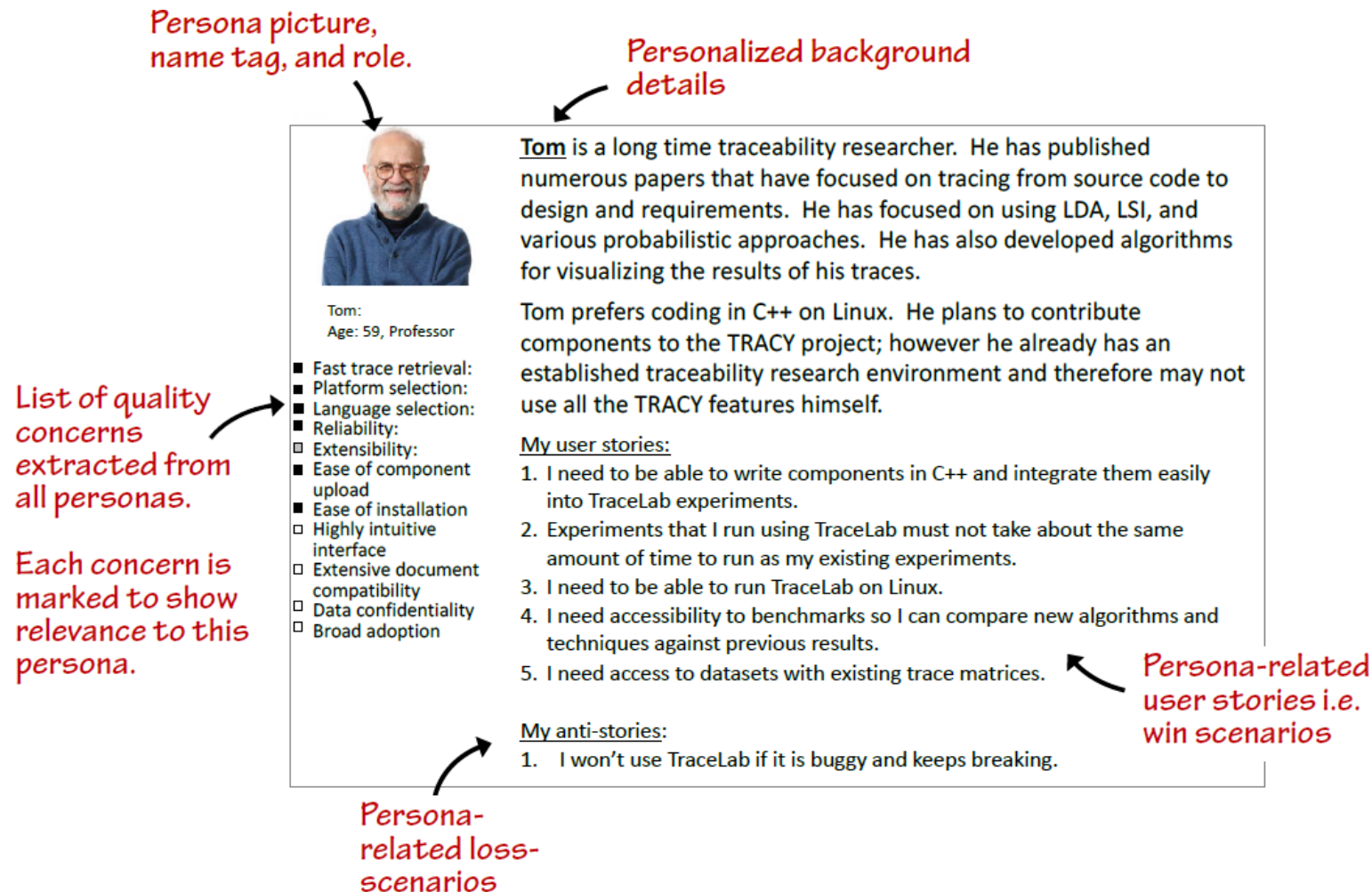
Persona-
related loss-
scenarios

ASP-Lite

Identify Personas

- Theory continued
 - Quality Concerns List
 - Quality Concerns Summary

Creation of a Persona in ASP-lite Part 2



ASP-Lite

The road to architectural design
decision

The road from personas to architectural design in ASP-lite Part I

- Each quality concern extracted from the user story is evaluated to identify ASR.
- The relevant user story for the concern is listed and connected to the persona
- ...Brainstorm for architectural solution based on identified user stories (The design approach is loosely following the Attribute Driven Design (ADD) process, with Quality Attribute Scenarios, and proposing and evaluation candidate solutions)
- Document architectural design decisions

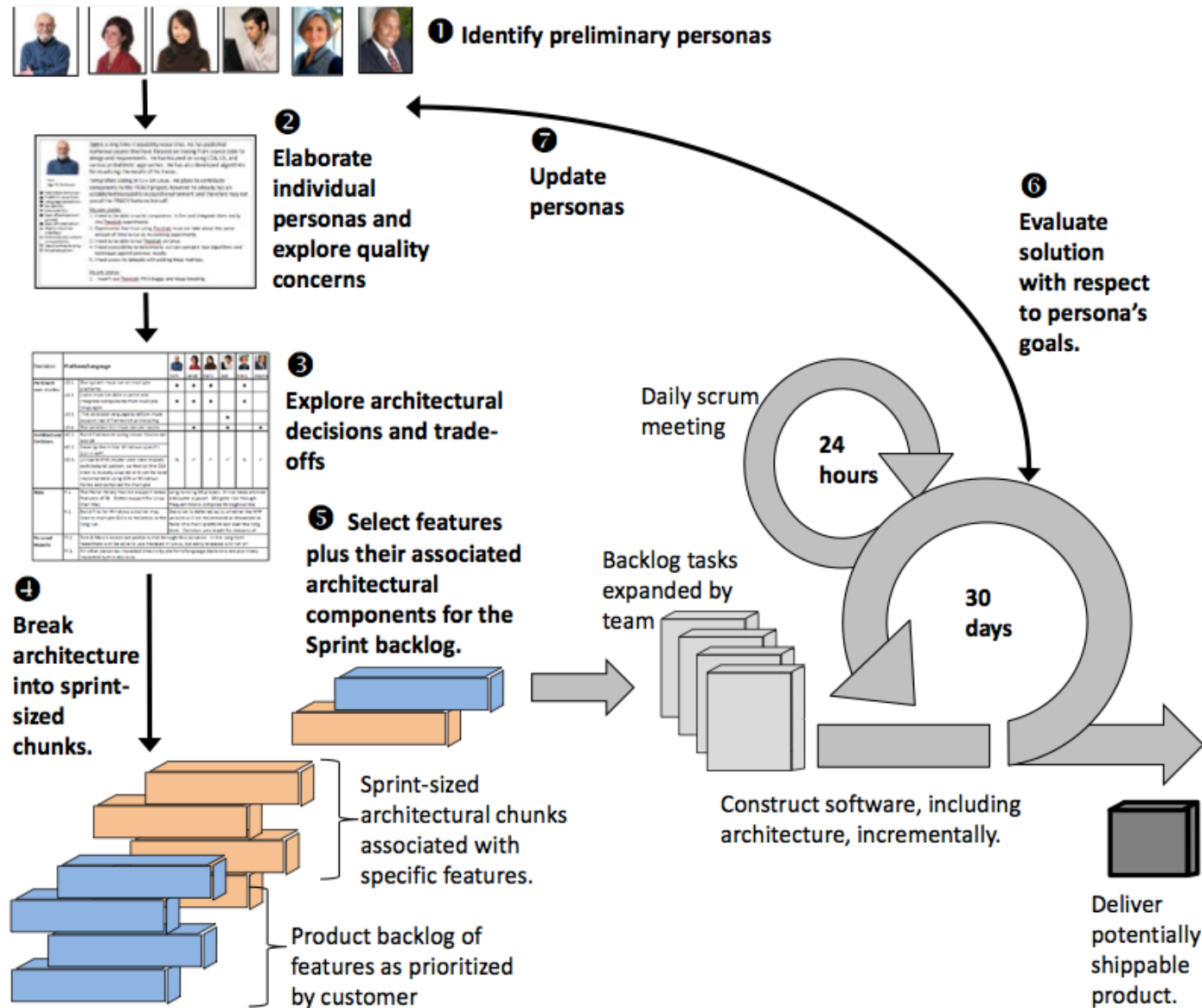
Decision:	Platform/Language							
Pertinent user stories:	US 1.	The system must run on multiple platforms	●	●	●		●	
	US 2.	Users must be able to write and integrate components from multiple languages	●	●	●		●	
	US 3.	The source language of each component must be invisible at runtime				●		
	US 4.	The selected language/platform must support rapid framework prototyping				●		
	US 5.	The selected GUI must deliver 'razzle dazzle'		●		●		●
Architectural Decisions	AD 1.	Build framework using Visual Studio.net and C#.						
	AD 2.	Develop the initial Windows-specific GUI in WPF.						
	AD 3.	Utilize MVVM (model view view model) architectural pattern, so that (a) the GUI View is loosely coupled and can be later implemented using GTK or Windows Forms and compiled for multiple platforms, and (b) the TraceLab engine can be compiled using Mono for porting to Linux and Mac environments.	½	✓	✓	✓	½	✓

The road from personas to architectural design in ASP-lite Part 2

- Identify architectural risks associated with the proposed solution and their mitigations.
- Consider and document impacts upon personas

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Risks	R 1.	The Mono library may not support latest features of C#. Better support for Linux than Mac.	Long running OS project. Initial tests showed adequate support. Mitigate risk through frequent Mono compiles throughout the project.					
	R 2.	Build first for Windows solution may lead to multiple GUIs to maintain in the long run.	Decision is deferred as to whether the WPF version will be maintained or discarded in favor of a multi-platform GUI over the long term.					
Personal Impacts	PI 1.	Tom & Mary's needs are partially met through this solution. In the long-term researchers will be able to use TraceLab in Linux, but early releases will run on Windows only.						
	PI 2.	All other personas impacted directly by platform/language decisions are positively impacted by this decision.						

Use of ASP-Lite in agile Projects



Questions
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Discussion

- We miss the use of the design decision template for capturing the rationale behind the decisions.
- Personas – it is vital to invite real stakeholders to participate in the workshop for capturing personas.